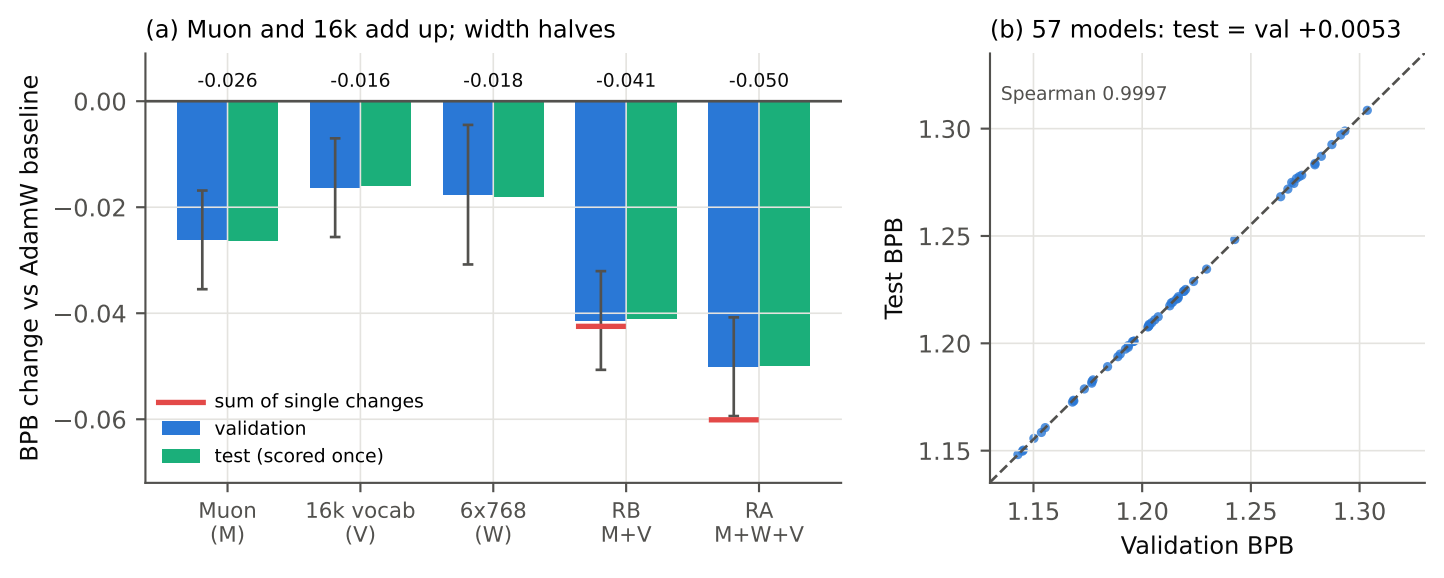
(a) Muon and 16k add up; width halves
BPB change vs AdamW baseline
0.00
−0.02
−0.04
−0.06
-0.026
-0.016
-0.018
-0.041
-0.050
sum of single changes
validation
test (scored once)
Muon (M)
16k vocab (V)
6x768 (W)
RB M+V
RA M+W+V
(b) 57 models: test = val +0.0053
Spearman 0.9997
Test BPB
1.30
1.25
1.20
1.15
1.15
1.20
1.25
1.30
Validation BPB